

Nucleophilic substitution reaction

Learning outcomes

At the end of this section you should be able to:

- (R3.4.1) Recognise the nucleophile in chemical reactions.
- (R3.4.2) Deduce equations with descriptions and explanations of the movement of electron pairs in nucleophilic substitution reactions.

When you learn and deduce the chemical equation between reactants and products (see [section R2.1.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/chemical-equations-id-43480/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/chemical-equations-id-43480/)), you need to rearrange reactant molecules to form new chemical bonds and produce product molecules. This will only happen when the reactant molecules share or transfer their electron pairs between them. Molecules that have a pair of electrons will be attracted to the positively charged regions on other molecules forming a new covalent bond. Do you think this will happen randomly? As you learn more about the fundamentals of electron pairs specifically in organic chemistry reactions, you will be able to understand the interaction of electrons between the reactant molecules.

Nucleophiles in chemical reactions

[Subtopic S2.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/) covers the covalent model, which involves a shared pair of electrons between two atoms. The formation of bonds occurs due to the tendencies of attracting electrons to gain stability by sharing their valence electrons. Additionally, a similar process happens in the [coordination bond formation](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/) but the sources of a pair of electrons are shared from the same atom (see [section S2.2.3 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/)).

Look at the molecules in **Figure 1**.

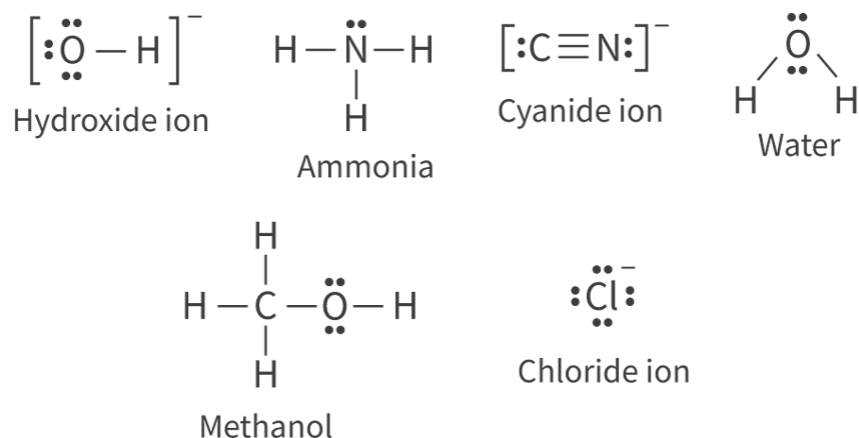


Figure 1. Several anions and neutral molecules.

More information for figure 1

The image is a diagram displaying the structures of various anions and neutral molecules. It includes the following chemical structures:

1. Hydroxide ion - depicted as "[O-H]-" with lone pairs on the oxygen.
2. Ammonia - depicted as "H-N-H" with three hydrogen atoms bonded to nitrogen.
3. Cyanide ion - depicted as "[C≡N]-" indicating a triple bond between carbon and nitrogen.
4. Water - depicted as "H-O-H" with lone pairs on oxygen.
5. Methanol - depicted as "H-C-O-H" with additional hydrogen atoms bonded to carbon.
6. Chloride ion - depicted as "[Cl]-" with lone pairs of electrons shown around the chlorine.

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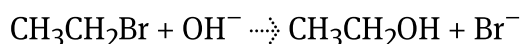
What do these molecules have in common and how do these molecules differ from each other?

You may notice that some of them are neutral (ammonia, water and methanol) and others are anions (hydroxide ion, chloride ion and nitrite ion). However, they all have at least one lone pair of electrons that are

available to form a new covalent bond with other atoms or molecules.

Anions with an extra electron cause them to more readily form a chemical bond with an electron-deficient species. Molecules that are shown with the non-bonding pairs of electrons above are examples of nucleophiles. A nucleophile is defined as an electron-rich species (neutral molecule or negatively charged ion) that has a tendency to donate a pair of non-bonding electrons to an electron-deficient species to form a new covalent bond.

Nucleophiles play an important role in many chemical reactions, including substitution reactions. For example, in the reaction between bromoethane and hydroxide ions.



What happens to a lone pair belonging to the hydroxide ion?

The non-bonding pairs of electrons on the hydroxide ion act as a nucleophile and are donated to the partially positively charged carbon atom of the C–Br bond. The electronegativity difference between carbon and bromine causes the carbon atom to become partially positively charged. As a result, the bromide ion is replaced by the hydroxide ion which produces ethanol and a bromide ion.

This process also occurs similarly to neutral species such as water.



Water as a nucleophile donates a pair of electrons to the partially positively charged carbon atom of the C–Br bond and replaces the bromine atom, forming ethanol and hydrogen bromide.

Common nucleophiles that are used in organic chemistry reactions: neutral molecules with a lone pair of electrons such as alcohols, water (H₂O), ammonia (NH₃) and negatively charged ions that have a lone pair of electrons such as halide ions (Cl[−], Br[−], I[−]), hydroxide ions (OH[−]) and cyanide ions (CN[−]).

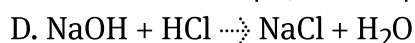
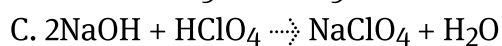
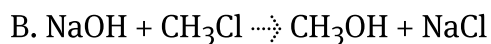
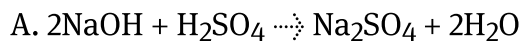
Study skills

The cyanide ion, CN[−], is a nucleophile that contains a triple bond between carbon and nitrogen atoms. This type of reaction is useful, as any halogenoalkanes reacting with cyanide ions will increase the length of the

carbon chain.

Worked example 1

From the reactions shown below,



which one involves NaOH acting as a nucleophile?

In reaction B, the hydroxide donates a lone pair of electrons to the partially positively charged carbon atom of the chloromethane, producing methanol and sodium chloride.

Reaction of A, C and D involves neutralisation where NaOH is acting as a base.

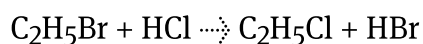


Exercise 1

Click a question to answer



1. Which of the following molecules acts as the nucleophile in the reaction below?



Check answers

Try the activity in **Interactive 1** to check your understanding of the key terms used in this section,

Drag the words into the correct boxes

Nucleophiles often participate in a variety of chemical reactions such as substitution reactions. A nucleophile is an _____ species and is typically negatively charged ion or _____ molecules that have a non-bonding pair of electrons. This available pair of electrons is donated to an _____ species and forms a new _____ bond. For example, a reaction between iodomethane (CH_3I) and a cyanide ion produces acetonitrile (CH_3CN) and an iodide ion. In this reaction the cyanide ion acts as a _____ attacking the _____ atoms in the iodomethane and replacing the iodine atoms.

covalent

electron-deficient

carbon

nucleophile

electron-rich

neutral

✔ Check

Interactive 1. Nucleophilic Substitution Reactions.

The movement of the electron pair in nucleophilic substitution reaction

In a chemical reaction, a nucleophile donates electron pairs to a positively charged region of a molecule (called an electrophile, which will be explained in more detail in [section R3.4.4 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrophilic-addition-reaction-id-46329/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrophilic-addition-reaction-id-46329/)) as shown in

Figure 2. This type of reaction is called a nucleophilic substitution reaction: Nu is nucleophilic, R is electrophile and X is the leaving group.

The movement of electron pairs in the chemical reactions is shown using 'curly arrows'. These are commonly used in organic reaction mechanisms to show the movement of electrons between reactants or between a nucleophile and a substrate, including when the electrons are pulled away to the leaving group that is being replaced.

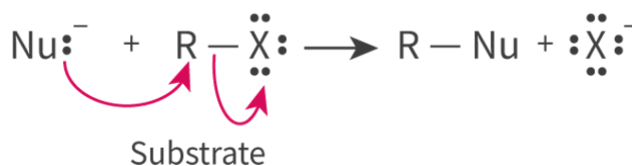


Figure 2. Equation of nucleophilic substitution reaction.

 More information for figure 2

The diagram illustrates a nucleophilic substitution reaction. On the left, the image shows a nucleophile (Nu:) interacting with a substrate (R-X) where R is a carbon chain and X is a leaving group. Curly arrows are used to indicate the movement of electron pairs: one arrow originates from the nucleophile pointing towards the carbon (R), while another curly arrow starts from R-X, showing the electrons moving away from the X, which is labeled as the leaving group. This results in the formation of a new compound on the right, R-Nu, where the nucleophile has replaced the leaving group X, now depicted with a pair of dots and a negative charge. The diagram explains the electron flow during the substitution process using the curly arrows to indicate bond formation and cleavage.

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These curly arrows help to illustrate how bonds are broken and formed during a chemical reaction. The destination of a pair of electrons is shown by the head of the curly arrow and the tail shows the source of a pair of electrons. The movement of a pair of electrons is represented by the direction of the arrow. For example, see the reaction of bromoethane and sodium hydroxide in **Figure 3**. Note that the sodium hydroxide dissociates in solution to produce the hydroxide ion, OH⁻, which then goes on to react with the bromoethane.



Figure 3. Nucleophilic substitution reaction of bromoethane with sodium hydroxide.

 More information for figure 3

The image is a diagram illustrating a nucleophilic substitution reaction. It shows hydroxide ion (:OH^-) reacting with bromoethane ($\text{C}_2\text{H}_5\text{-Br}$). The hydroxide ion acts as a nucleophile, indicated by a curly arrow pointing from :OH^- to the carbon atom in $\text{C}_2\text{H}_5\text{-Br}$, which is partially positively charged. This leads to the formation of ethanol ($\text{C}_2\text{H}_5\text{-OH}$) and the bromide ion (Br^-). The bromine atom in bromoethane is shown as the leaving group with a pair of electrons, indicated by another curly arrow pointing from the bond between carbon and bromine toward the bromine atom, which is then stabilized as Br^- .

[Generated by AI]

The hydroxide ion from sodium hydroxide acts as a nucleophile and attacks carbon because it is partially positively charged, and replaces bromine as the leaving group (**Figure 4**). A lone pair of hydroxide ions is donated to the carbon atom forming a new bond between carbon and hydroxide to produce ethanol. The leaving group is considered more stable than the nucleophile as it is less reactive and able to retain its electronic configuration by pulling back its electrons when they are being replaced by the nucleophiles.

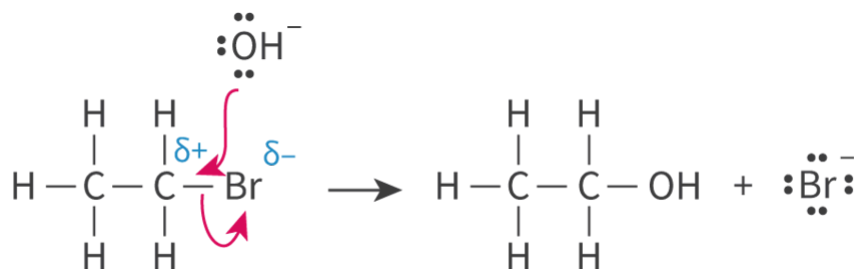


Figure 4. The (hydroxide ion) nucleophile attacks carbon on bromoethane.

 More information for figure 4

The image is a chemical diagram illustrating a nucleophilic substitution reaction. It shows the molecular structures before and after the reaction. On the left side, bromoethane is depicted, with a partially positively charged carbon atom (δ^+) bonded to bromine, which has a partial negative charge (δ^-). A hydroxide ion (:OH^-) is shown approaching the carbon atom, with an arrow indicating the direction of the attack. The arrow starts from the lone pair on the hydroxide ion and points towards the carbon-bromine bond, illustrating the substitution process. After the reaction arrow, on the right side,

ethanol (H-C-C-OH) is displayed as the product along with bromine ion (:Br⁻), indicating the completion of the reaction. The diagram captures the flow of electrons and the change in bonding resulting from the substitution.

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In general, a nucleophile is more likely to play a part in a chemical reaction depending on its reactivity. The more reactive the nucleophile, the more easily it donates electrons. This is usually affected by the electronegativity of the nucleophile, the use of solvent for the reaction that takes place and the electrophile that the nucleophile is reacting with.

The role of nucleophiles in chemical reactions typically occurs by following one of several mechanisms that will be explained in more detail in the HL section R3.4.9–10 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-mechanism-hl-id-46331/>). This depends on the nature of both the donor and acceptor of a pair of electrons including the reaction conditions.

Study skills

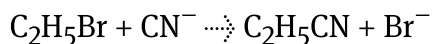
Nucleophiles do not attack positively charged molecules randomly. Their interaction occurs depending on the specific and predictable manner referring to the chemical properties of both molecules that donate and accept the electrons. In this way, you can predict the possible outcomes.

Try **Worked example 1** and the exercise question on the role of nucleophiles in chemical reactions.

Worked example 1

Deduce the chemical equation for the reaction between bromoethane and the cyanide ion. Explain their interaction and the movement of electron pairs in that reaction.

The chemical equation for the reaction between bromoethane and the cyanide ion is as follows:



The movement of a lone pair of electrons on the cyanide ion approaches the partially positive carbon atom of the C–Br bond. The carbon–bromine bond then breaks, causing the bromine atom to depart from the molecule as a leaving group taking both electrons from the carbon–bromine bond, forming a bromine ion. A new bond between the carbon atom and the cyanide ion is formed, producing propanenitrile.



Exercise 2

Click a question to answer



1. Which of the following reactions correctly shows the electrons movement when methanol as a nucleophile reacts with chloromethane?



Check answers

Activity

- **IB learner profile attribute:** Communicator
- **Approaches to learning:** Communication — Negotiate ideas and knowledge
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Pair activity

Your task

Pair up with another student and write the products of the chemical equations below. Include the electron pairs of the nucleophile, substrate and the products, as well as the curly arrow to show the movement of the electrons (refer to **Figure 2**).

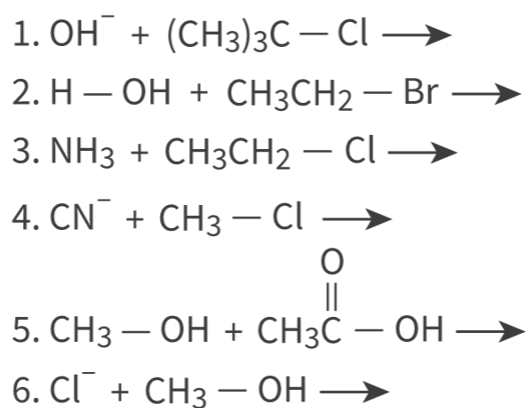


Figure 5. Write the products of these chemical equations.

 More information for figure 5

The image contains six numbered chemical equations featuring various reactions requiring the identification of reaction products, electron pairs, and electron movement through a curly arrow notation.

1. Equation 1: $\text{OH}^- + (\text{CH}_3)_3\text{C} - \text{Cl} \rightarrow$
2. Equation 2: $\text{H} - \text{OH} + \text{CH}_3\text{CH}_2 - \text{Br} \rightarrow$
3. Equation 3: $\text{NH}_3 + \text{CH}_3\text{CH}_2 - \text{Cl} \rightarrow$
4. Equation 4: $\text{CN}^- + \text{CH}_3 - \text{Cl} \rightarrow$
5. Equation 5: $\text{CH}_3 - \text{OH} + \text{CH}_3\overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{OH} \rightarrow$ (with a double-bonded oxygen on top of the second carbon)
6. Equation 6: $\text{Cl}^- + \text{CH}_3 - \text{OH} \rightarrow$

Each reaction includes reactants like hydroxide, water, ammonia, cyanide ion, and chloride ion, paired with various organic substrates like tert-butyl chloride and ethanol, requiring prediction of the products and demonstrating the arrow mechanisms to account for electron shifts.

[Generated by AI]

When you are finished, discuss with your partner what kind of difficulties you encountered to complete this activity? Is there any specific concept you need to highlight?

The mark scheme is available [here](#).

